Lab Diagram and Topology Description: Automating VLAN Configuration Using Python and NETCONF

Network Topology Diagram

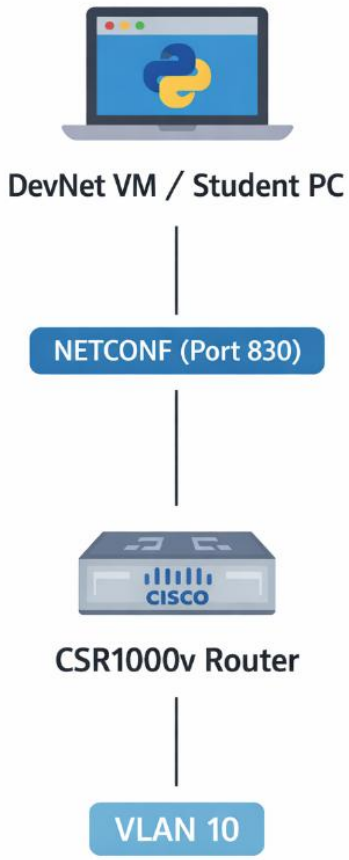

Figure 1. Network Topology for VLAN Automation Lab

This diagram illustrates the communication between the DevNet VM and the CSR1000v router using NETCONF over TCP port 830.

Topology Description

The lab environment is straightforward, containing two devices, a DevNet virtual machine and a Cisco CSR1000v router. The DevNet VM will be considered the student workstation that runs Python automation scripts. The CSR1000v is the network device being configured.

DevNet VM communicates with the router using NETCONF protocol over TCP port 830. Python scripts use this connection to push configuration to the router and pull configuration from the router for verification.

The DevNet VM directly connects to the router via its management IP address. This address will change depending on whether you finish the lab in the NDG NetLab environment or use a locally hosted DevNet VM.

Communication Details

The following communication occurs within the topology:

- Protocol: NETCONF over SSH
- Port: TCP 830
- Source Device: DevNet VM (Python automation client)
- Destination Device : Cisco CSR1000v router
- Login: username/password (cisco/cisco or admin/admin based on your environment)

Device Roles

1) **DevNet Virtual Machine**

- Runs Python scripts (`vlan_config.py, verify_vlan.py`)
- Acts as the automation client
- Initiates NETCONF connections to the router

2) **Cisco CSR1000v Router**

- Receives configuration via NETCONF
- Implements VLAN subinterface configuration
- Provides configuration data for verification

Design Considerations

This topology has been kept simple so students can concentrate on automation rather than trying to understand complex topology. With only one router and one automation client students will easily be able to see how configurations are pushed and validated through code.

NETCONF allows for a more standardized method of communication which helps promote best practices with network automation.

Environment Flexibility

This topology can be deployed in:

- NDG NetLab DevNet environment
- Locally hosted DevNet VM with CSR1000v router

While the IP addressing and credentials may vary between environments, the topology and automation workflow remain the same.